

Syed Abdullah Ahmed

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Dear Professor Usman and Dr. Ali,

I'm excited to apply for the Research at York (RAY) position with the Humanitarian Water Engineering Lab. As a Computer Science student at York University with a background in Python development, I see this role as the perfect way to bring my skills into work that has a real humanitarian impact.

Growing up in Singapore for eight years exposed me to a blend of cultures and perspectives, which shaped how I approach collaboration, listening first, adapting, and finding common ground. That's carried through to my work. During the pandemic, I co-founded Rozgari, a non-profit platform connecting daily-wage workers to clients through an online and SMS booking system. We raised 1 million PKR, onboarded 25+ service providers, and averaged five bookings per day within a year. Proof that technology can directly improve lives and make a positive impact on society. That experience was a big part of what inspired me to pursue my degree in Computer Science.

Most recently, I interned at Alethea AI, where I worked extensively with Django (Python Framework), FastAPI, Postman, and DBeaver to build and optimize backend systems. I implemented CRUD operations, designed and queried databases, and created efficient API endpoints. I also led a code cleanup initiative to remove obsolete environment variables and APIs, which improved maintainability, and performed blackbox testing on the company's decentralized app to identify and help resolve critical bugs. This role taught me how to approach problems systematically, manage complex datasets, and ensure accuracy and reliability.

I first learned about the Safe Water Optimization Tool (SafeH2O/SWOT) through my conversations with Professor Usman, and I was immediately intrigued. The way it uses field water quality data and modeling to generate real-time, site-specific chlorination guidance is both fascinating and practical. It's a perfect example of research turned into actionable solutions, something I'd be proud to contribute to!

I'm drawn to this position not just because of the technical fit, but because of the mission: turning complex research into tools and training that help humanitarian professionals make better decisions in the field. With my technical background, research mindset, and eagerness to learn, I'm confident I can add value to the lab's ongoing projects while growing as a researcher.

Thank you for considering my application. I'd love the opportunity to bring my skills, adaptability, and energy to the Humanitarian Water Engineering Lab.

Best,
Abdullah